Chapter 8 — Physical Optics and Gravitational Waves

Complete Notes · PECTAA Physics 11 · Board-style longs + conceptual shorts

Topic 1 — Polarization of light

Q.1 Explain polarization of light and describe its types and production by polaroids.

Polarization is the restriction of the vibrations of a transverse wave to a particular direction. It proves that light is a transverse wave, because longitudinal waves cannot be polarized. In plane-polarized light the electric field vibrates in one plane; in circularly and elliptically polarized light the tip of the electric field vector rotates in a circle or ellipse.

Plane-polarized light can be produced by passing ordinary light through a polaroid, which transmits only the component of vibration along its transmission axis. A second polaroid used to examine the light is called an analyzer. When the two polaroids are crossed at 90°, no light is transmitted, which confirms the transverse nature of light.

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Important Conceptual Short Questions — Topic 1

1. Why can longitudinal waves not be polarized?

In a longitudinal wave the vibrations are along the direction of travel, so there is no perpendicular vibration direction to restrict. Polarization is therefore possible only for transverse waves.

2. How do crossed polaroids show that light is transverse?

A polaroid passes only one vibration direction. When a second polaroid is crossed at 90°, it blocks that direction completely. This would not happen for longitudinal waves, so light must be transverse.

3. Why is the intensity reduced when ordinary light passes through a polaroid?

Ordinary light contains vibrations in all perpendicular directions. The polaroid transmits only the component along its axis, so about half the intensity is removed.

Topic 2 — Brewster’s law, Malus’s law and optical activity

Q.2 State Brewster’s law and Malus’s law and explain optical activity.

When light is reflected from a transparent surface at the polarizing angle θ_p, the reflected light is plane-polarized. Brewster’s law states that $\tan \theta_p = n$, where n is the refractive index of the medium. At this angle the reflected and refracted rays are 90° apart.

Malus's law gives the intensity transmitted by an analyzer: $I = I_0 \cos^2\theta$, where I_0 is the incident polarized intensity and θ is the angle between the transmission axes. The amplitude passed is $A_0 \cos \theta$, and intensity is proportional to amplitude squared. Optically active substances such as quartz and sugar solution rotate the plane of polarization of light passing through them.

Important Conceptual Short Questions — Topic 2

1. Why does Malus's law contain cos squared rather than cos?

The analyzer passes the amplitude component $A_0 \cos \theta$. Since intensity is proportional to the square of amplitude, the transmitted intensity is $I_0 \cos^2\theta$.

2. How is Brewster's angle related to the refractive index?

At the polarizing angle the reflected and refracted rays are perpendicular. Applying Snell's law with this condition gives $\tan \theta_p = n$.

3. Why are polaroids used in sunglasses and sky photography?

Polaroids block the polarized glare reflected from surfaces such as water or road, and they darken the polarized light scattered by the sky, improving contrast and reducing unwanted light.

4. How are polaroids used in stress analysis of materials?

A transparent model under stress becomes birefringent and rotates the plane of polarization differently in different regions. Viewed between crossed polaroids, the stressed regions appear as coloured fringes, showing where the stress is greatest.

Topic 3 — Gravitational waves and their types

Q.3 Describe gravitational waves and explain their four basic types.

Gravitational waves are ripples in space-time produced by accelerating masses, predicted by Einstein in 1916. They travel outward at the speed of light and stretch and compress objects perpendicular to their direction of travel. They pass through matter almost unchanged, but their amplitude is extremely small, so they are very difficult to detect.

The four basic types are continuous waves from a steadily spinning massive object such as a neutron star; compact binary inspiral waves from two orbiting compact objects such as black holes or neutron stars; burst waves from violent events such as supernovae; and stochastic waves, a random background from many weak distant sources.

Textbook check (p. 177): the first detection is linked to colliding neutron stars. The 2015 signal came from two black holes; the first neutron-star detection was in 2017. Also p. 179 prints 10^{-21} to 10^{-22} in metres, but this range is strain, a dimensionless ratio; the actual LIGO arm change is about 10^{-18} m.

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Important Conceptual Short Questions — Topic 3

1. Why can we not detect gravitational waves from everyday moving objects?

The masses and accelerations of everyday objects are far too small to produce waves of measurable amplitude. Only very massive, rapidly accelerating objects such as merging black holes produce detectable waves.

2. How do gravitational waves affect an object as they pass?

They produce tidal stretching and compressing perpendicular to the direction of travel, alternately lengthening and shortening the object by an extremely small amount.

3. Why are gravitational waves useful for studying the universe?

They pass through matter almost unabsorbed, so they carry information from distant violent events such as black-hole mergers that light cannot reveal.

Topic 4 — Detection of gravitational waves by interferometers

Q.4 Explain how a laser interferometer detects gravitational waves.

A laser interferometer such as LIGO splits a laser beam into two perpendicular arms, reflects the beams from mirrors and recombines them. In the absence of a gravitational wave the arm lengths are adjusted so that the recombined beams interfere destructively and no light reaches the detector. A passing gravitational wave changes the two arm lengths by different tiny amounts, disturbing the interference and producing a detectable signal.

LIGO has two 4 km arms in the USA and made the first detection on 14 September 2015, announced in 2016. Virgo near Pisa has 3 km arms. The 2017 Nobel Prize in Physics was awarded to Rainer Weiss, Barry Barish and Kip Thorne; Nergis Mavalvala of MIT, of Pakistani origin, was part of the team.

Textbook check (p. 181): a light year is printed as 9.5 billion km. It is about 9.5 trillion km.

Important Conceptual Short Questions — Topic 4

1. Why are very long arms used in LIGO?

The change in length produced by a gravitational wave is proportional to the arm length. Longer arms give a larger absolute change and make the tiny strain easier to detect.

2. How does interference make the interferometer sensitive to tiny length changes?

The recombined beams are set to cancel. A very small change in one arm changes the phase difference and spoils the cancellation, so even a minute length change produces a measurable light signal.

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3. Why are two detectors such as LIGO and Virgo used together?

A real gravitational wave should appear in both detectors with the correct time delay, while local disturbances appear in only one. Using two detectors confirms the signal and helps locate its source.

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